

One-Page Summary Report

AnalystLab Africa Data Analytics Internship Program

Week 1: Data Cleaning & Exploratory Data Analysis (EDA)

Dataset: Online Retail Transactions

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Cleaning Challenges Encountered

During the data cleaning process, several data quality issues were identified and resolved to prepare the dataset for analysis. The **Description** column contained **1,454** missing values (0.27%), and the affected records were removed because the missing product descriptions prevented product identification. The **CustomerID** column contained **135,080** missing values (24.93%). These records were retained because they represented valid sales transactions, although they were excluded from customer-level analysis.

A total of **5,268** duplicate records were identified and removed to eliminate redundant transactions. Column names were standardized to lowercase, and the **InvoiceDate** column was converted from **object** to **datetime** format to support accurate time-based analysis. Data validation also identified **9,725** records with negative quantities, **1,058** records with zero or negative unit prices, and **9,251** cancelled transactions. These invalid records were removed to ensure that the final dataset contained only valid sales transactions suitable for analysis.

Key EDA Findings

- **PAPER CRAFT, LITTLE BIRDIE** was the best-selling product, with **80,995** units sold.
- The **United Kingdom** generated the highest revenue (approximately **£9.0 million**).
- **November 2011** recorded the highest monthly sales revenue, while **February 2011** recorded the lowest.
- **WHITE HANGING HEART T-LIGHT HOLDER** was the most frequently purchased product.
- **Customer 14646** generated the highest total sales revenue.

Top Insights

- The United Kingdom is the company's largest and most profitable market.
- Home décor and gift products showed the strongest customer demand.
- Sales increased significantly during the final quarter of the year, indicating seasonal demand.
- A small number of high-value customers contributed a substantial share of total revenue.
- Revenue-based analysis provided deeper business insights than sales quantity alone.

